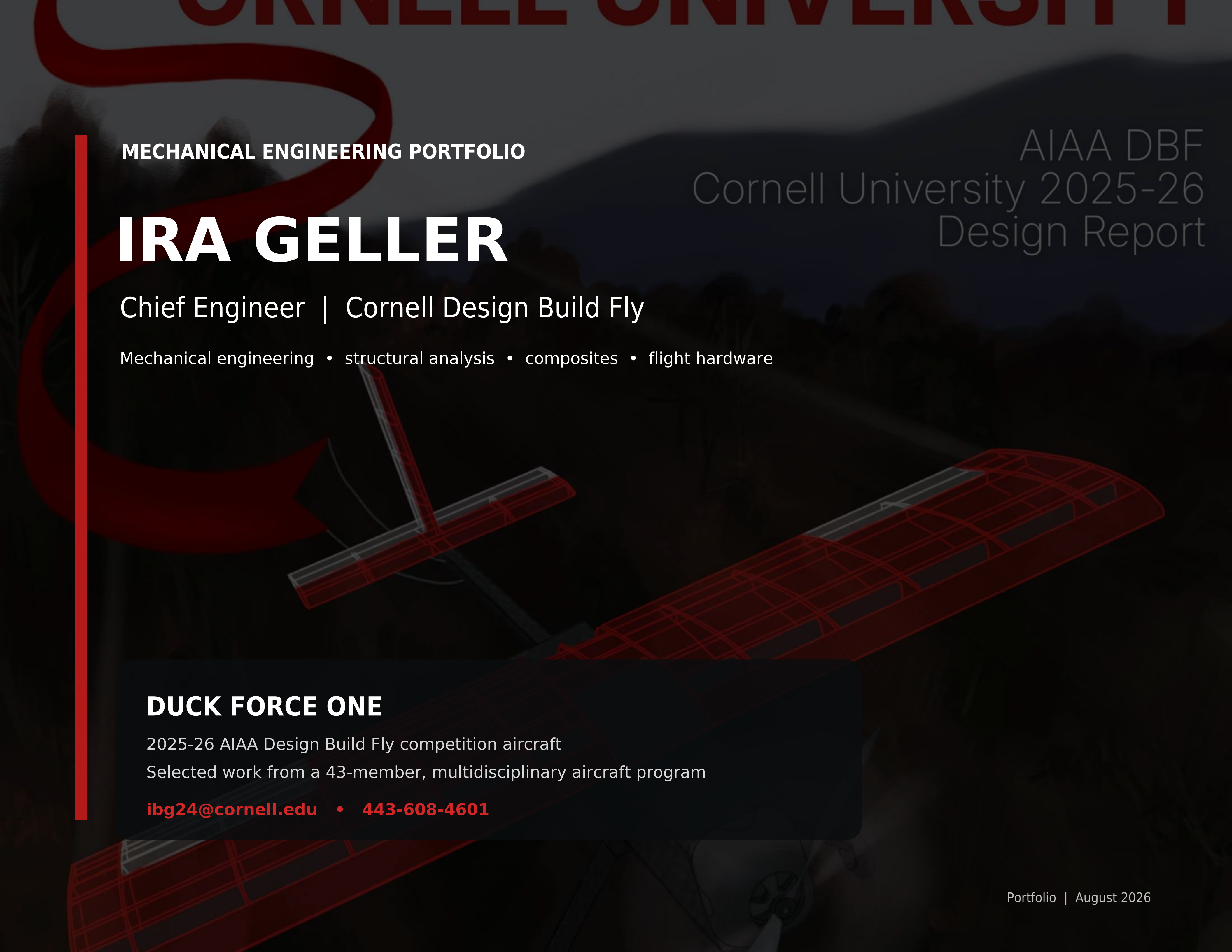


MECHANICAL ENGINEERING PORTFOLIO

IRA GELLER

Chief Engineer | Cornell Design Build Fly

Mechanical engineering • structural analysis • composites • flight hardware

DUCK FORCE ONE

2025-26 AIAA Design Build Fly competition aircraft

Selected work from a 43-member, multidisciplinary aircraft program

ibg24@cornell.edu • 443-608-4601

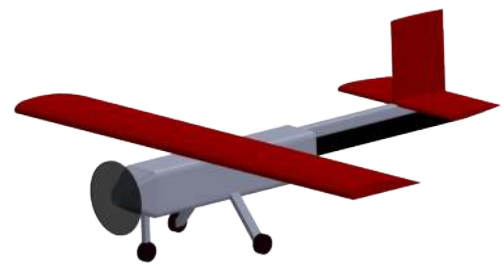
Turning requirements into a flight-ready system

ROLE / CHIEF ENGINEER

I set technical direction and led system integration from competition requirements through flight testing.

MY CONTRIBUTION

- Owned tradeoffs among performance, reliability, manufacturability, and schedule.
- Led design reviews and integration across aerodynamic, propulsion, and mechanical systems.



DUCK FORCE ONE

High-wing, single-motor bush-plane configuration carrying passengers and cargo, with an in-flight deployable 150 in banner.

43
TEAM SIZE
undergraduates

7.37 lb
M3 MTOW
predicted

SYSTEMS ENGINEERING LOOP

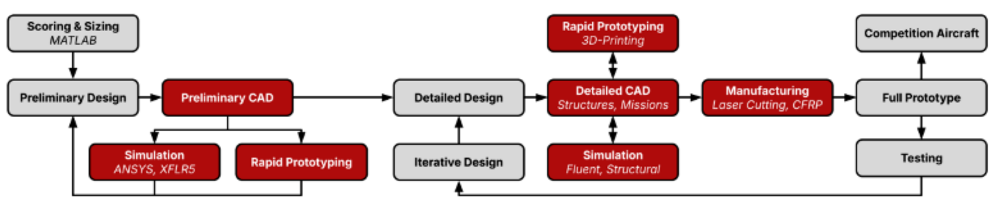
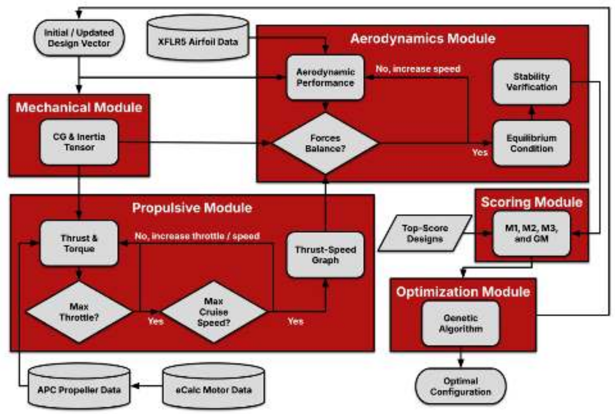


Figure 4.1.1: Design process and procedures

Requirements and scoring drove analysis; analysis drove CAD; fabricated hardware and test data drove the next design iteration.

Model the system. Verify the hardware.

MULTIDISCIPLINARY DESIGN OPTIMIZATION



Aerodynamics, mass properties, propulsion, scoring, and a genetic algorithm were coupled in MATLAB.

OPTIMIZATION OUTPUT

500k+
DESIGNS
evaluated

31 vars
DESIGN VECTOR
geometry + config

150 in
BANNER
AR = 5.00

The model revealed two high-scoring design families. The selected concept prioritized Ground Mission and banner performance while maintaining feasible stability, thrust, endurance, and structural margins.

DESIGN-SPACE EXPLORATION

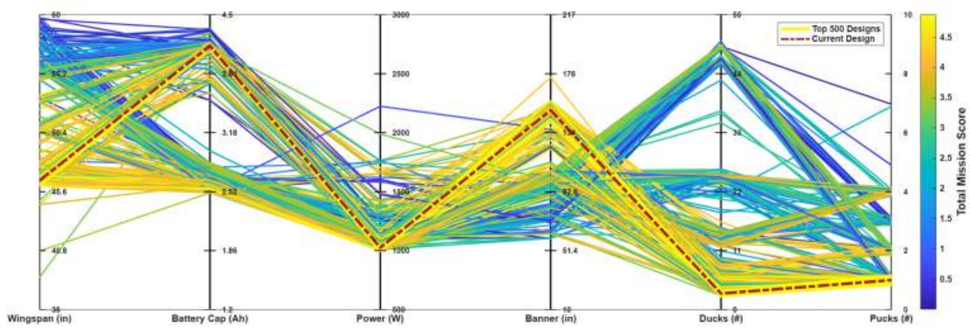
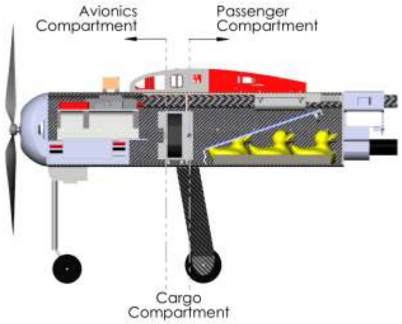


Figure 4.3.1: Desing parameter scoring study

The current design (red dashed) was selected from feasible candidate designs after trading wingspan, battery capacity, power, banner length, and payload.

Design for load paths, integration, and buildability

CFRP MONOCOQUE FUSELAGE



Two 3-ply clamshells, internal bulkheads, and local load-distribution plates created a lightweight structure with dedicated avionics, cargo, and passenger compartments.

FoS 5.6 extreme combined load case

WING ASSEMBLY FEA

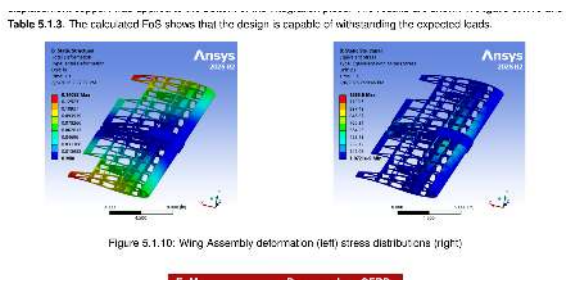


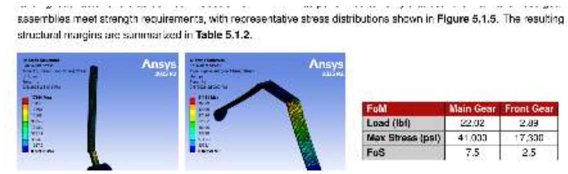
Table 5.1.3. The calculated FoS shows that the design is capable of withstanding the expected loads.

Figure 5.1.10: Wing Assembly deformation (left) stress distributions (right)

A 32 lbf distributed load represented the 3.5-g design condition. The rib-and-stringer wing used a square CFRP spar and a 3D-printed integration structure.

0.14 in maximum basswood deflection

LANDING GEAR


assemblies meet strength requirements, with representative stress distributions shown in Figure 5.1.5. The resulting structural margins are summarized in Table 5.1.2.

FoM	Main Gear	Front Gear
Load (lbf)	22.02	2.89
Max Stress (psi)	41,000	17,300
FoS	7.5	2.5

Carbon-fiber main gear and a steerable aluminum nose gear were sized for a 7.4 lb MTOW and a 3.5 landing load factor.

7.5 / 2.5 main / nose gear FoS

ANALYSIS-DRIVEN DESIGN DECISIONS

- Rejected wing endplates: marginal aerodynamic benefit did not justify added mass, complexity, and build time.
- Rejected landing-gear fairings: predicted drag reduction was offset by the additional weight.
- Used battery position as movable ballast to maintain acceptable CG across three mission configurations.
- Checked stability, control authority, servo torque, motor cooling, crosswind capability, and structural margins before release to fabrication.

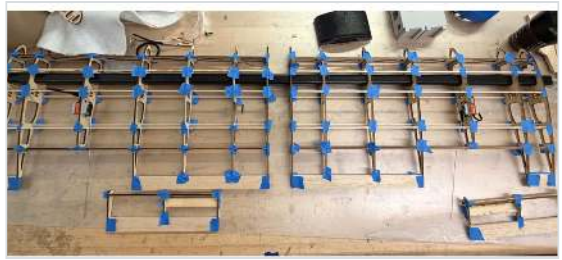
TOOLS SOLIDWORKS • ANSYS Mechanical / ACP / Fluent • XFLR5 • MATLAB

CAD, FEA, CFD, flight mechanics, stability, mass properties, propulsion, and mission scoring

04 / BUILD, TEST, ITERATE

Hardware results closed the loop

FABRICATION + TESTING



MEASURE. LEARN. REDESIGN

35.21 → 15.27 s
GROUND MISSION
iteration 1 to iteration 4

26.70 lbf
WING SPAR LOAD TEST
0.94 in deflection • validated spar design

TAKEOFF + LANDING
FLIGHT TEST
sustained, controlled flight

FAILURE BECAME A DESIGN INPUT

The first flight lost transmitter connection after carbon-fiber signal attenuation. Post-flight inspection identified the antenna-placement issue; the next iteration added an external receiver on the opposite side of the fuselage and completed sustained flight with a controlled landing.

WHAT THIS PROJECT TAUGHT ME

High-performance hardware comes from tight feedback loops: define the requirement, make the tradeoff explicit, build the simplest testable system, learn from real hardware, and iterate quickly without losing technical rigor.

TECHNICAL OWNERSHIP
SYSTEM INTEGRATION
TEST-DRIVEN ITERATION